



JKBOSE 2026

CLASS 11TH

CONIC SECTIONS

MATHS

LECTURE-6

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TOPICS TO BE COVERED

1

Exercise Discussion 10.3

2

Related Problems

$$c = \sqrt{a^2 - b^2}$$

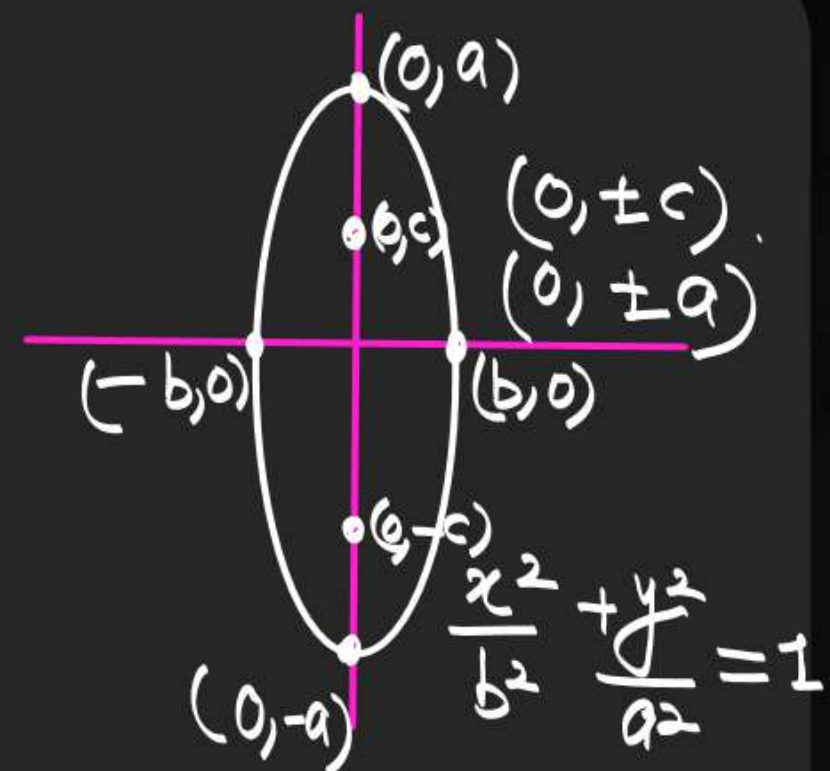
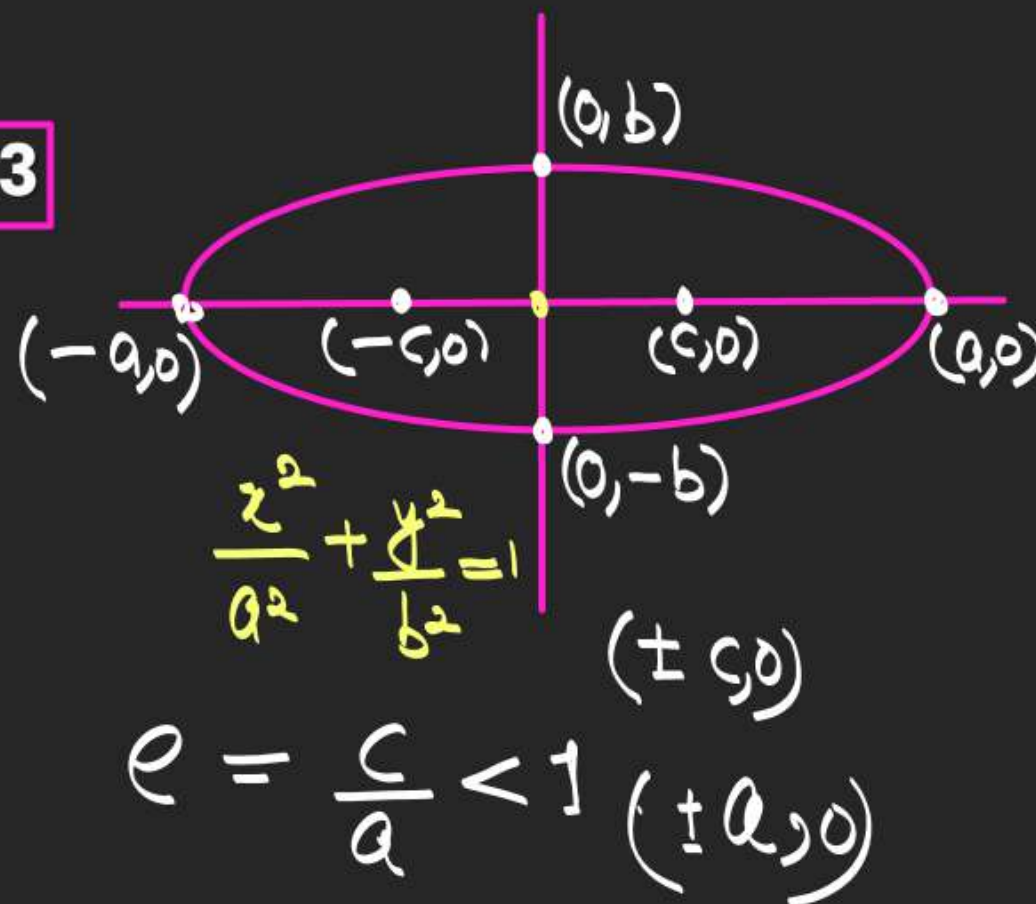
$$c^2 = a^2 - b^2$$

$$e^2 = \frac{c^2}{a^2}$$

$$a^2 e^2 = a^2 - b^2$$

$$b^2 = a^2 - a^2 e^2$$

$$\Rightarrow b^2 = a^2(1 - e^2)$$



Latus rectum

Suppose that the given ellipse is:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad \text{--- (1)}$$

Since (c, l) lies on (1), So therefore we get;

$$\frac{c^2}{a^2} + \frac{l^2}{b^2} = 1$$

$$\Rightarrow \frac{a^2 - b^2}{a^2} + \frac{l^2}{b^2} = 1$$

$$\Rightarrow \frac{\cancel{a^2} - b^2}{\cancel{a^2}} + \frac{l^2}{b^2} = 1$$

$$\Rightarrow \cancel{1} - \frac{b^2}{a^2} + \frac{l^2}{b^2} = \cancel{1}$$

$$\Rightarrow -\frac{b^2}{a^2} + \frac{l^2}{b^2} = 0$$

$$\Rightarrow \frac{l^2}{b^2} = \frac{b^2}{a^2}$$

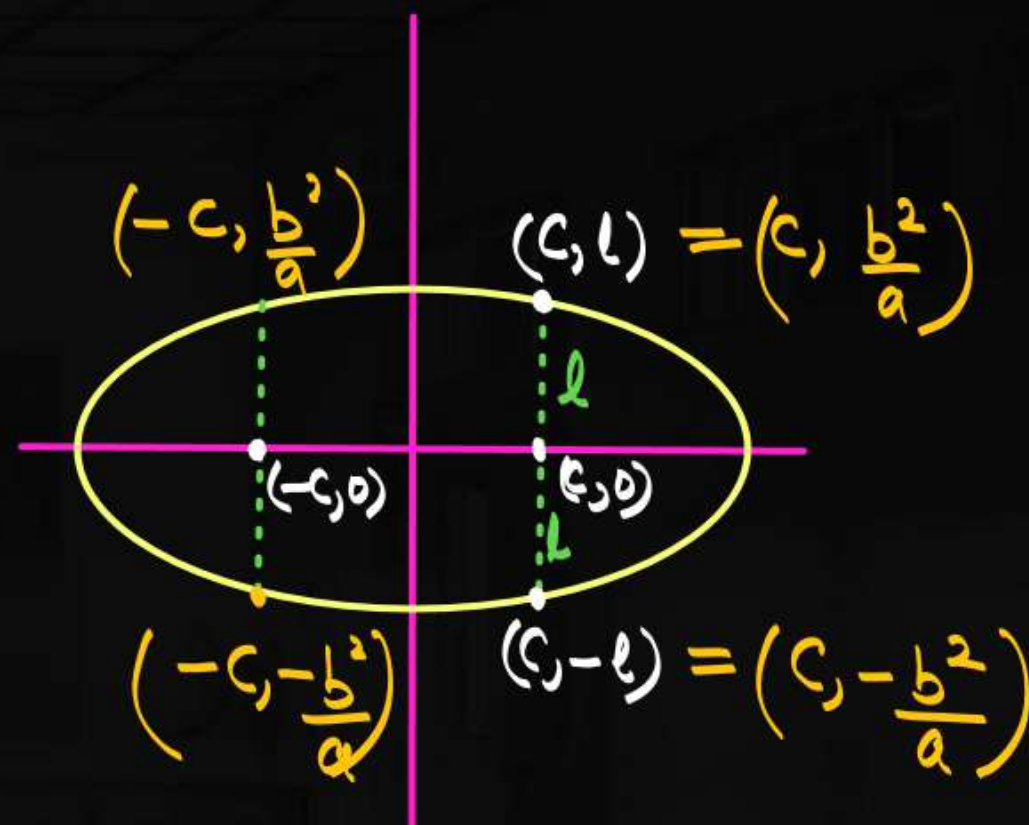
$$\Rightarrow l^2 = \frac{b^4}{a^2}$$

$$\Rightarrow \boxed{l = \frac{b^2}{a}}$$

So, therefore the length of latus Rectum
 $= 2l$

$$= \frac{2b^2}{a}$$

Hence the result.



EXAMPLE

Find the coordinates of the foci, the vertices, the length of major axis, the minor axis, the eccentricity and the latus rectum of the ellipse

$$\frac{x^2}{25} + \frac{y^2}{9} = 1$$

(Sol) Given ellipse is:

$$\frac{x^2}{25} + \frac{y^2}{9} = 1$$

$$\Rightarrow \frac{x^2}{5^2} + \frac{y^2}{3^2} = 1$$

$$\text{So, } a = 5, b = 3$$

$$\Rightarrow c = \sqrt{a^2 - b^2}$$

$$\Rightarrow c = \sqrt{25 - 9} = \sqrt{16}$$

$$\Rightarrow c = 4$$

Therefore Co-ordinates of foci = $(\pm 4, 0)$

Co-ordinates of vertices = $(\pm 5, 0)$

length of major axis = $2a = 10$

length of minor axis = $2b = 6$

eccentricity, $e = \frac{4}{5}$

Also, length of latus rectum

$$= \frac{2b^2}{a}$$

$$= \frac{2 \times 9}{5}$$

$$= \frac{18}{5} \text{ units.}$$

Hence the result.

EXAMPLE

Find the coordinates of the foci, the vertices, the lengths of major and minor axes and the eccentricity of the ellipse $9x^2 + 4y^2 = 36$.

(Sol) Given ellipse is:

$$9x^2 + 4y^2 = 36$$

$$\Rightarrow \frac{9x^2}{36} + \frac{4y^2}{36} = 1$$

$$\Rightarrow \frac{x^2}{4} + \frac{y^2}{9} = 1$$

$$\Rightarrow \frac{x^2}{2^2} + \frac{y^2}{3^2} = 1$$

$$\text{So, } a = 3, b = 2$$

$$\Rightarrow c = \sqrt{9 - 4}$$

$$\Rightarrow c = \sqrt{5}$$

Therefore the Co-ordinates of foci = $(0, \pm \sqrt{5})$

Co-ordinates of the vertices = $(0, \pm 3)$

length of major axis = $2a = 6$

length of minor axis = $2b = 4$

eccentricity, $e = \frac{c}{a} = \frac{\sqrt{5}}{3}$

Also, length of latus Rectum

$$= \frac{2b^2}{a} = \frac{2 \times 4}{3}$$

$$= \frac{8}{3} \text{ units.}$$

Hence the result.

EXAMPLE

Find the equation of the ellipse, with major axis along the x -axis and passing through the points $(4, 3)$ and $(-1, 4)$.

(Sol) Suppose that the given ellipse is:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad \text{--- (1)}$$

Since $(4, 3)$ and $(-1, 4)$ lies on (1), so we get;

$$\frac{16}{a^2} + \frac{9}{b^2} = 1 \quad \& \quad \frac{1}{a^2} + \frac{16}{b^2} = 1$$

$$\Rightarrow 16u + 9v = 1 \quad \text{--- (1)} \quad \& \quad u + 16v = 1$$

$$\text{or } u = 1 - 16v \quad \text{--- (2)}$$

$$\text{Where } u = \frac{1}{a^2} \quad \& \quad v = \frac{1}{b^2}$$

Using (2) in (1), we get;

$$16(1 - 16v) + 9v = 1$$

$$\Rightarrow 16 - 256v + 9v = 1$$

$$\Rightarrow -247v = -15$$

$$\Rightarrow v = \frac{15}{247}$$

$$\text{So, } u = 1 - \frac{16 \times 15}{247}$$

$$\Rightarrow u = \frac{247 - 240}{247}$$

$$\Rightarrow u = \frac{7}{247}$$

$$\text{So, } \frac{1}{a^2} = \frac{7}{247}, \quad \frac{1}{b^2} = \frac{15}{247}$$

$$\Rightarrow a^2 = \frac{247}{7}, \quad b^2 = \frac{247}{15}$$

Therefore from (1), we get

$$\left(\frac{x^2}{\frac{247}{7}}\right) + \left(\frac{y^2}{\frac{247}{15}}\right) = 1$$

$$\Rightarrow 7x^2 + 15y^2 = 247 \quad \text{Which is the required ellipse. Hence the result.}$$

QUESTION (NCERT EXAMPLER):

The equation of the ellipse whose centre is at the origin and the x -axis, the major axis, which passes through the points $(-3, 1)$ and $(2, -2)$ is.

Homework.

1 $5x^2 + 3y^2 = 32$

2 $3x^2 + 5y^2 = 32$

3 $5x^2 - 3y^2 = 32$

4 $3x^2 + 5y^2 + 32 = 0$

QUESTION (NCERT EXAMPLER):

The length of the Latus rectum of the ellipse $3x^2 + y^2 = 12$ is:

- 1 ☒ 4
- 2 ☒ 3
- 3 ☒ 8
- 4 ☒ $\frac{4}{\sqrt{3}}$

(Sol) Given ellipse is: $3x^2 + y^2 = 12$.

$$\Rightarrow \frac{3x^2}{12} + \frac{y^2}{12} = 1$$

$$\Rightarrow \frac{x^2}{4} + \frac{y^2}{12} = 1$$

So, here $a^2 = 12$, $b^2 = 4$.

$$\Rightarrow a = \sqrt{12} = 2\sqrt{3}, \quad b = 2$$

$$\Rightarrow \text{length of latus Rectum} = \frac{2b^2}{a} = \frac{2 \times 4}{2\sqrt{3}} = \frac{4}{\sqrt{3}}$$

QUESTION (NCERT EXAMPLER):

If e is the eccentricity of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, ($a < b$), then

Semi-minor axis = a

Semi-major axis = b

1 $b^2 = a^2(1 - e^2)$

2 $a^2 = b^2(1 - e^2)$

3 $a^2 = b^2(e^2 - 1)$

4 $b^2 = a^2(e^2 - 1)$

$$* e = \frac{c}{b} = \frac{b^2 - a^2}{b^2}$$

$$\Rightarrow b^2 e^2 = b^2 - a^2$$

$$\Rightarrow a^2 = b^2(1 - e^2)$$

QUESTION (NCERT EXAMPLER):

The equation of the circle passing through the foci of the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ and having centre at $(0, 3)$ is:

1 $x^2 + y^2 - 6y - 7 = 0$ ✓

2 $x^2 + y^2 - 6y + 7 = 0$ ✗

3 $x^2 + y^2 - 6y - 5 = 0$ ✗

4 $x^2 + y^2 - 6y + 5 = 0$ ✗

(Sol) let the required circle be
 $(x-0)^2 + (y-3)^2 = r^2$
 $\Rightarrow x^2 + (y-3)^2 = r^2$ — (1)

Here $c = \sqrt{16-9} = \sqrt{7}$
 $\Rightarrow$ foci = $(\pm\sqrt{7}, 0) = (\sqrt{7}, 0), (-\sqrt{7}, 0)$
 Since $(\sqrt{7}, 0)$ lies on (1), we get;

$7 + (0-3)^2 = r^2$

$\Rightarrow r^2 = 7+9$

$\Rightarrow r^2 = 16$

So, the required equation of the circle is:

$x^2 + (y-3)^2 = 16$

$\Rightarrow x^2 + y^2 + 9 - 6y - 16 = 0$

$\Rightarrow x^2 + y^2 - 6y - 7 = 0$

QUESTION

If the eccentricity of the two ellipses $\frac{x^2}{169} + \frac{y^2}{25} = 1$ and $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ are equal, then $\frac{a}{b}$ is equal to :

~~1~~ $\frac{5}{13}$

~~2~~ $\frac{6}{13}$

☒ 3 $\frac{13}{5}$

~~4~~ $\frac{13}{6}$

(Sol) Here $e_1 = \frac{\sqrt{169-25}}{13} = \frac{12}{13}$

$$e_2 = \frac{c}{a}$$

It's given that $e_1 = e_2$.

$$\Rightarrow \frac{12}{13} = \frac{c}{a}$$

$$\Rightarrow \left(\frac{\sqrt{a^2 - b^2}}{a} \right)^2 = \left(\frac{12}{13} \right)^2$$

$$\Rightarrow \frac{a^2 - b^2}{a^2} = \frac{144}{169}$$

$$\Rightarrow 1 - \frac{b^2}{a^2} = \frac{144}{169}$$

$$\Rightarrow \frac{b^2}{a^2} = 1 - \frac{144}{169} = \frac{169 - 144}{169}$$

$$\Rightarrow \left(\frac{b}{a} \right)^2 = \frac{25}{169}$$

$$\Rightarrow \frac{b}{a} = \sqrt{\frac{25}{169}} = \frac{5}{13}$$

$$\Rightarrow \boxed{\frac{a}{b} = \frac{13}{5}}$$

QUESTION

An ellipse, with the foci at $(0, 2)$ and $(0, -2)$ and minor axis of length 4, passes through which of the following points?

~~1~~ $(2, \sqrt{2})$

~~2~~ $(2, 2\sqrt{2})$

~~3~~ $(\sqrt{2}, 2)$

~~4~~ $(1, 2\sqrt{2})$

(Sol) Suppose that $\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$ — ①
be the given ellipse.

Here $2b = 4 \Rightarrow b = 2$

$c = 2$

$\Rightarrow c^2 = 4$

$\Rightarrow a^2 - b^2 = 4$

$\Rightarrow a^2 - 4 = 4$

$\Rightarrow a^2 = 8$

$\Rightarrow a = 2\sqrt{2}$

Therefore from ①, we get;

$$\frac{x^2}{4} + \frac{y^2}{8} = 1$$

$\Rightarrow 2x^2 + y^2 = 8$

EXERCISE – 10.3

QUESTION

In each of the Exercises 1 to 9, find the coordinates of the foci, the vertices, the length of major axis, the minor axis, the eccentricity and the length of the latus rectum of the ellipse.

$$\frac{x^2}{36} + \frac{y^2}{16} = 1 \quad \text{HP/W}$$

QUESTION

In each of the Exercises 1 to 9, find the coordinates of the foci, the vertices, the length of major axis, the minor axis, the eccentricity and the length of the latus rectum of the ellipse.

$$\frac{x^2}{4} + \frac{y^2}{25} = 1 \quad \text{H/w}$$

$$(0, \pm c)$$

$$(0, \pm a)$$

QUESTION

In each of the Exercises 1 to 9, find the coordinates of the foci, the vertices, the length of major axis, the minor axis, the eccentricity and the length of the latus rectum of the ellipse.

$$\frac{x^2}{16} + \frac{y^2}{9} = 1$$

hp/w

QUESTION

In each of the Exercises 1 to 9, find the coordinates of the foci, the vertices, the length of major axis, the minor axis, the eccentricity and the length of the latus rectum of the ellipse.

$$\frac{x^2}{25} + \frac{y^2}{100} = 1$$

hp/w

QUESTION

In each of the Exercises 1 to 9, find the coordinates of the foci, the vertices, the length of major axis, the minor axis, the eccentricity and the length of the latus rectum of the ellipse.

$$\frac{x^2}{49} + \frac{y^2}{36} = 1$$

HP/W

QUESTION

In each of the Exercises 1 to 9, find the coordinates of the foci, the vertices, the length of major axis, the minor axis, the eccentricity and the length of the latus rectum of the ellipse.

$$\frac{x^2}{100} + \frac{y^2}{400} = 1$$

HP/W

QUESTION

In each of the Exercises 1 to 9, find the coordinates of the foci, the vertices, the length of major axis, the minor axis, the eccentricity and the length of the latus rectum of the ellipse.

$$36x^2 + 4y^2 = 144$$

$$\frac{36x^2}{144} + \frac{4y^2}{144} = 1$$

$$\Rightarrow \frac{x^2}{4} + \frac{y^2}{36} = 1$$

$$(0, \pm c)$$

$$(0, \pm a)$$

$$a = 6$$

$$b = 2$$

$$c = \sqrt{a^2 - b^2}$$

QUESTION

In each of the Exercises 1 to 9, find the coordinates of the foci, the vertices, the length of major axis, the minor axis, the eccentricity and the length of the latus rectum of the ellipse.

$$16x^2 + y^2 = 16$$

H/W

QUESTION

In each of the Exercises 1 to 9, find the coordinates of the foci, the vertices, the length of major axis, the minor axis, the eccentricity and the length of the latus rectum of the ellipse.

$$4x^2 + 9y^2 = 36$$

HP/W

QUESTION

In each of the following Exercises 10 to 20, find the equation for the ellipse that satisfies the given conditions:

Vertices $(\pm 5, 0)$, foci $(\pm 4, 0)$

$$a = 5, \quad c = 4$$

$$c^2 = 16$$

$$a^2 - b^2 = 16$$

$$25 - 16 = b^2$$

$$b^2 = 9$$

$$\frac{x^2}{25} + \frac{y^2}{9} = 1$$

QUESTION

In each of the following Exercises 10 to 20, find the equation for the ellipse that satisfies the given conditions:

Vertices $(0, \pm 13)$, foci $(0, \pm 5)$

a ✓ c

$$c = 5$$

$$c^2 = 25$$

$$a^2 - b^2 = 25$$

QUESTION

In each of the following Exercises 10 to 20, find the equation for the ellipse that satisfies the given conditions:

Vertices $(\pm 6, 0)$, foci $(\pm 4, 0)$

QUESTION

In each of the following Exercises 10 to 20, find the equation for the ellipse that satisfies the given conditions:

Ends of major axis $(\pm, 3, 0)$, ends of minor axis $(0, \pm 2)$

HP/W

QUESTION

In each of the following Exercises 10 to 20, find the equation for the ellipse that satisfies the given conditions:

Ends of major axis $(0, \pm\sqrt{5})$, ends of minor axis $(\pm 1, 0)$

$$a = \sqrt{5}, b = 1$$

$$a^2 = 5, b = 1$$

QUESTION

In each of the following Exercises 10 to 20, find the equation for the ellipse that satisfies the given conditions:

Length of major axis 26, foci $(\pm 5, 0)$

$2a = 26$
 $a = 13, \quad c^2 = 25$
 $a^2 - b^2 = c^2$
 $13^2 - b^2 = 25$
 $169 - b^2 = 25$
 $-b^2 = 25 - 169$
 $-b^2 = -144$
 $b^2 = 144$

QUESTION

In each of the following Exercises 10 to 20, find the equation for the ellipse that satisfies the given conditions:

Length of minor axis 16, foci $(0, \pm 6)$

LP/n:

QUESTION

In each of the following Exercises 10 to 20, find the equation for the ellipse that satisfies the given conditions:

Foci $(\pm 3, 0)$, $a = 4$

QUESTION

In each of the following Exercises 10 to 20, find the equation for the ellipse that satisfies the given conditions:

$b = 3$, $c = 4$, centre at the origin; foci on the x axis.

$$a^2 - b^2 = 16.$$

$$a^2 = 25.$$

QUESTION

In each of the following Exercises 10 to 20, find the equation for the ellipse that satisfies the given conditions:

Centre at $(0, 0)$, major axis on the y -axis and passes through the points $(3, 2)$ and $(1, 6)$.

$$\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$$

QUESTION

In each of the following Exercises 10 to 20, find the equation for the ellipse that satisfies the given conditions:

Major axis on the x -axis and passes through the points $(4, 3)$ and $(6, 2)$.

H/W

Syed Fakhra



Tawkeer Amin Sofi



Aksha De.



Thank You

love —→ Freedom.
Attachment —→ Cage.